

INTEGRATED PROJECT - II

Theme : Smart City Provision

Multi-Priority Emergency Vehicle By Using RF Beacon System With Automatic Reset Logic

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Motivation: Lives Lost Due to Traffic Delays

Traffic congestion significantly delays emergency vehicles such as ambulances, fire trucks, and police vehicles. These delays can lead to the loss of many lives that could otherwise be saved with faster response times.

Key Statistics:

- In **India**, around **24,000 patients die every day** due to delays in receiving timely medical help, many caused by traffic congestion and slow ambulance response.
- India records about **1.5 lakh road accident deaths every year**, and many victims die before reaching the hospital because of delayed emergency care.
- Studies show that **about 30% of accident deaths could be prevented** if ambulances reach the hospital faster.

Real-Life Scenario Example

An ambulance stuck at a traffic signal for **5–10 minutes** can mean the difference between life and death for patients suffering from **heart attacks, strokes, or severe injuries**.

✓ **This motivates the need for a Multi-Priority Emergency Vehicle System** that automatically clears traffic signals and provides priority to emergency vehicles.



PROBLEM STATEMENT:

Real-World Issue

- Emergency vehicles (Ambulance, Fire Engine, Police) face delays at traffic signals.
- Manual traffic control causes response time delay.
- Accidents and loss of life occur due to late arrival of emergency services.

Why It Is Important

- Every second matters in medical emergencies.
- Faster response = Higher survival rate.
- Smart traffic control can save lives.

Existing Limitations

- Manual override systems.
- No priority differentiation.
- No automatic reset after emergency passes.
- Complex and costly smart traffic systems.

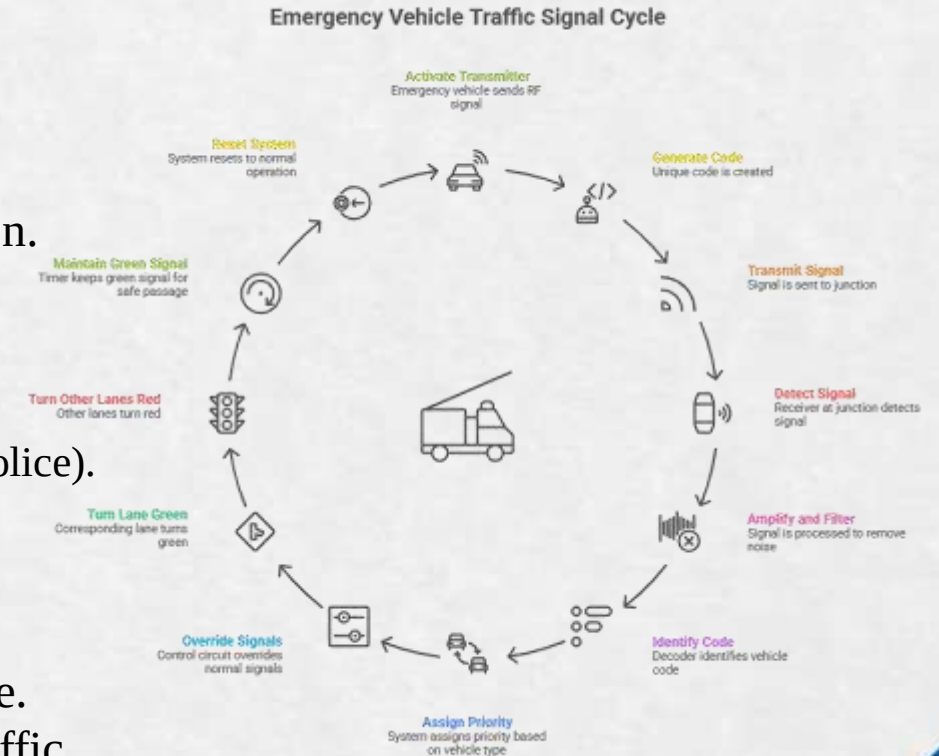
OBJECTIVES:

- Detect approaching emergency vehicles.
- Assign different priority levels.
- Automatically control traffic signals.
- Reduce emergency response time.
- Improve road safety.
- Implement cost-effective smart solution.

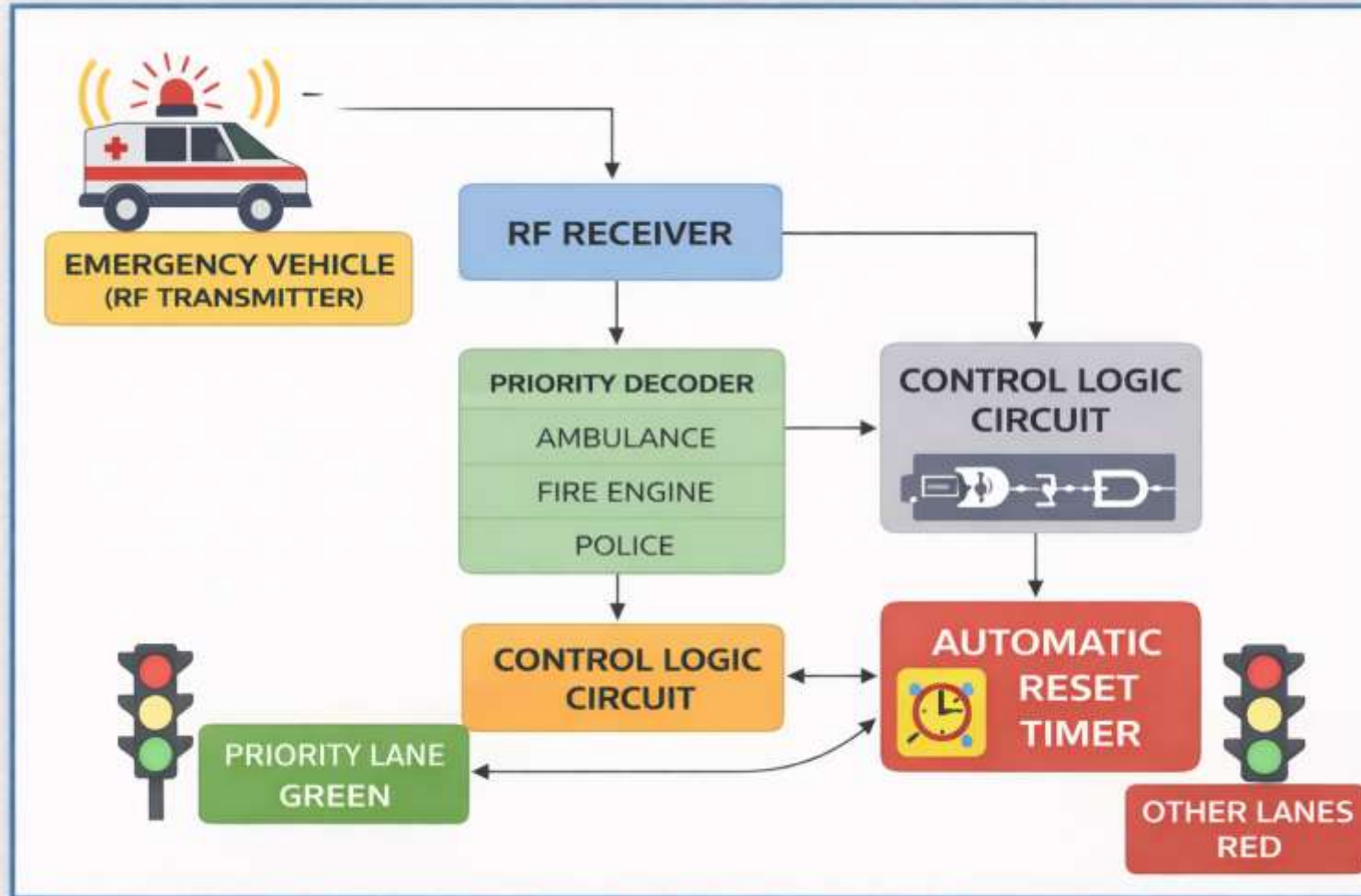
METHODOLOGY / APPROACH:

Process flow:

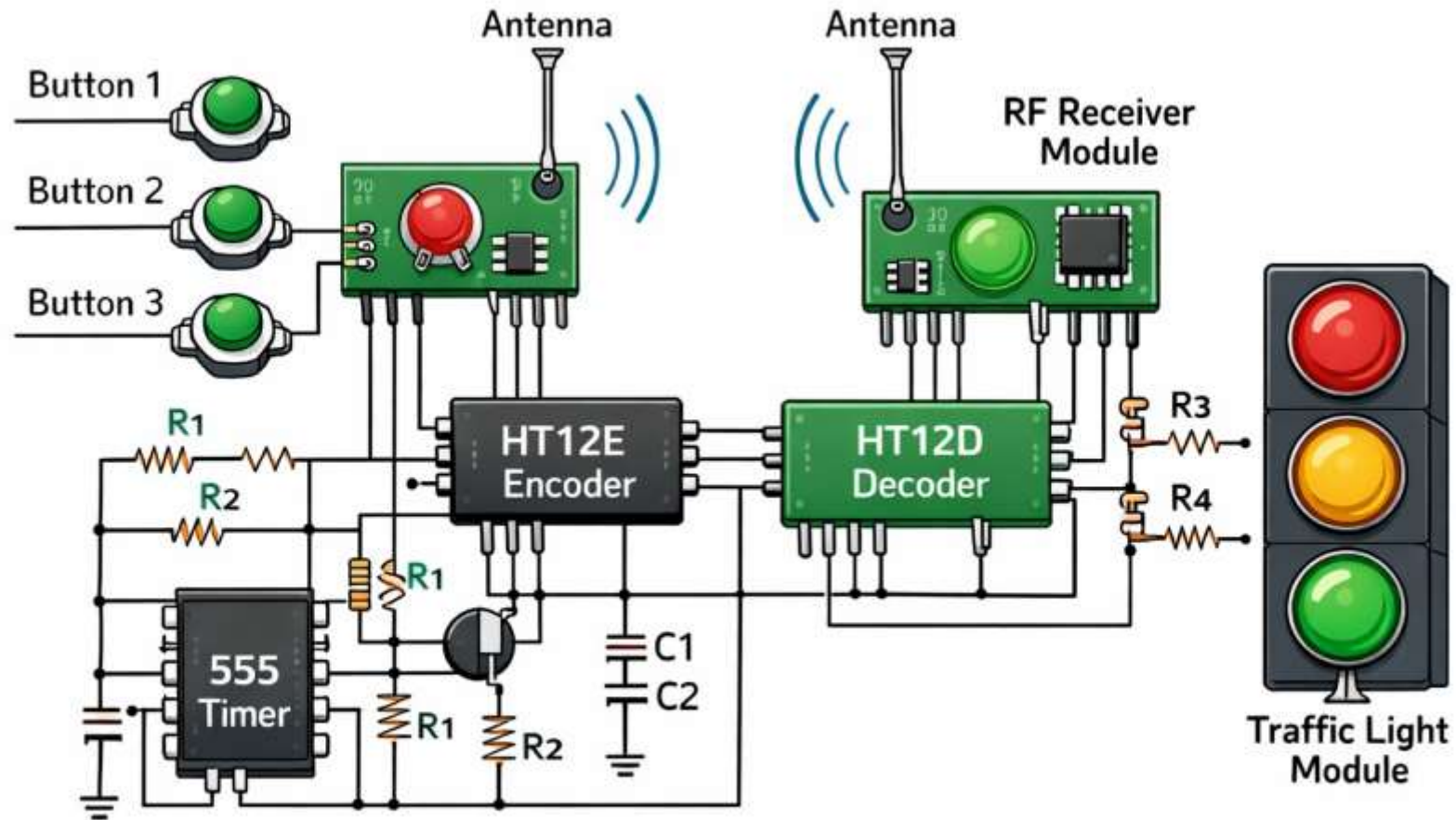
- The emergency vehicle activates its RF transmitter.
- A unique coded RF signal is generated based on vehicle type.
- The RF signal is transmitted wirelessly toward the traffic junction.
- The RF receiver at the signal detects the incoming signal.
- The received signal is amplified and filtered to remove noise.
- The decoder verifies and identifies the emergency vehicle code.
- The system determines the priority level (Ambulance > Fire > Police).
- The control logic circuit overrides the normal traffic sequence.
- The corresponding lane signal turns GREEN.
- All other lanes are switched to RED to clear the path.
- A preset timer maintains the green signal for safe vehicle passage.
- After the delay, the system automatically resets to the normal traffic cycle.



BLOCK DIAGRAM:



Circuit Diagram:



ANALYTICAL WORK:

The proposed system is based on the following core concepts:

RF Communication Principle – Wireless transmission of coded signals between emergency vehicle and traffic signal.

Priority Encoding Logic – Assigning priority levels to different emergency vehicles.

Timer-Based Reset Mechanism – Ensuring automatic restoration of normal operation.

Mathematical & Analytical Calculations

555 Timer Delay Calculation

The time delay for automatic reset is calculated using:

$$T = 1.1 \times R \times C$$

Where:

T = Time delay (seconds)

R = Resistance (Ohms)

C = Capacitance (Farads)

This determines how long the green signal remains active.

Priority Logic Condition

- If multiple signals are received:
- Ambulance → Highest priority output
- Fire Engine → Medium priority
- Police → Lower priority
- Logic implementation ensures only one highest priority lane becomes GREEN.

🎯 Analytical Outcome

- Accurate timing control achieved.
- Reliable signal detection ensured.
- Proper priority-based switching validated.
- Automatic reset verified.

Simulation Circuit:

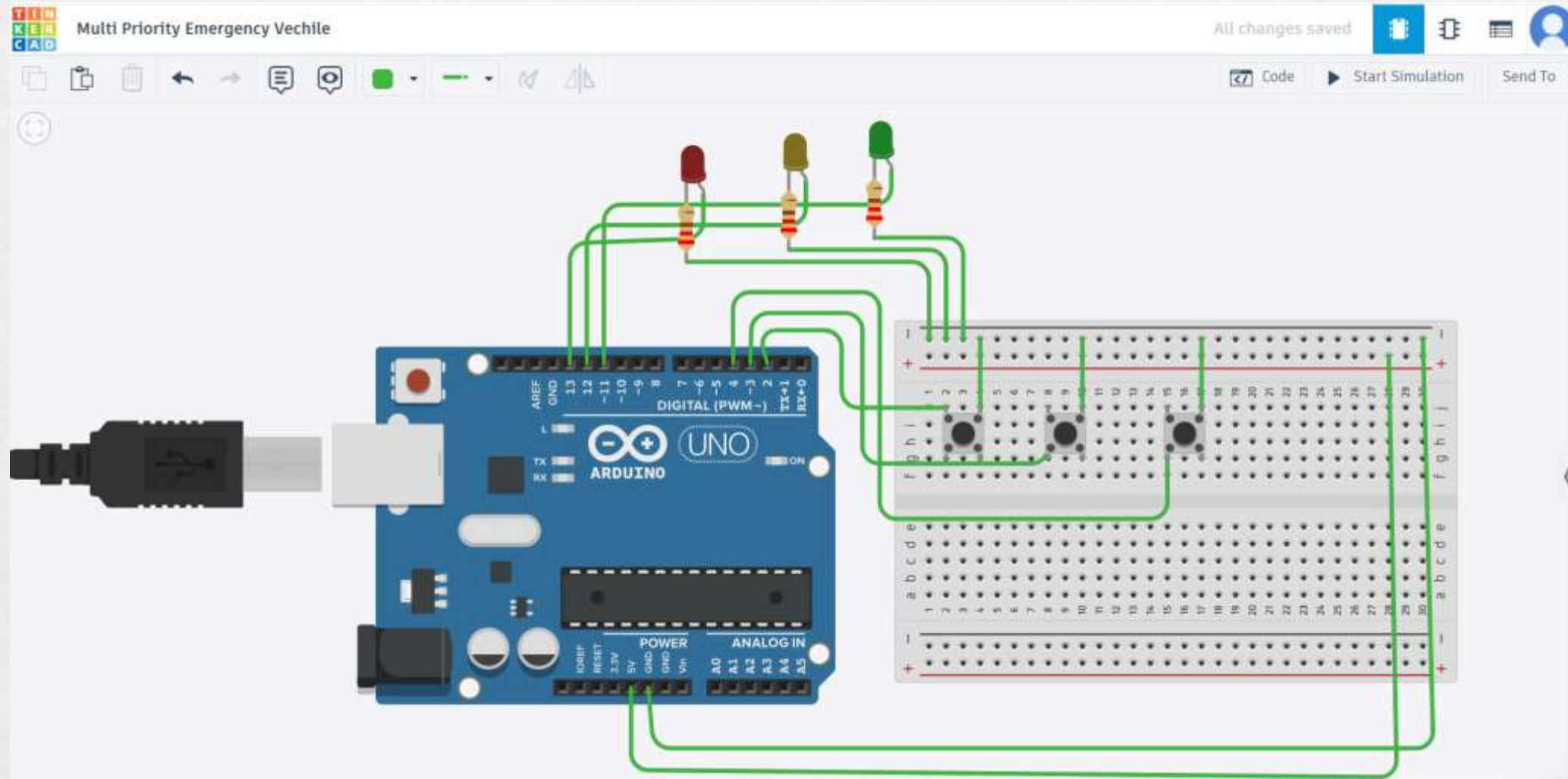
We explored different circuit simulation software for this project Such as

- Multi-Sim
- Wokwi
- EasyEda
- Everycircuit



But many required hardware components were not available in those tools. Therefore, we used Tinkercad because it provides essential components like Arduino, push buttons, LEDs, and resistors. Tinkercad also allows easy circuit design and simulation, which helped us successfully implement and test the project.

Tinker CAD Simulation :



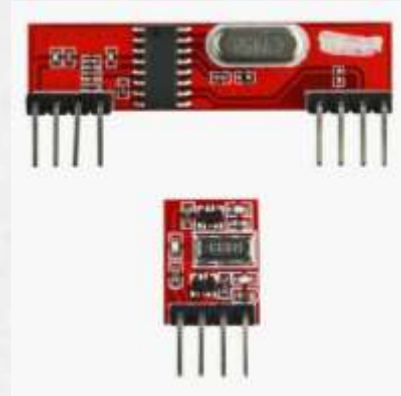
Hardware Implementation:

Hardware Components Used

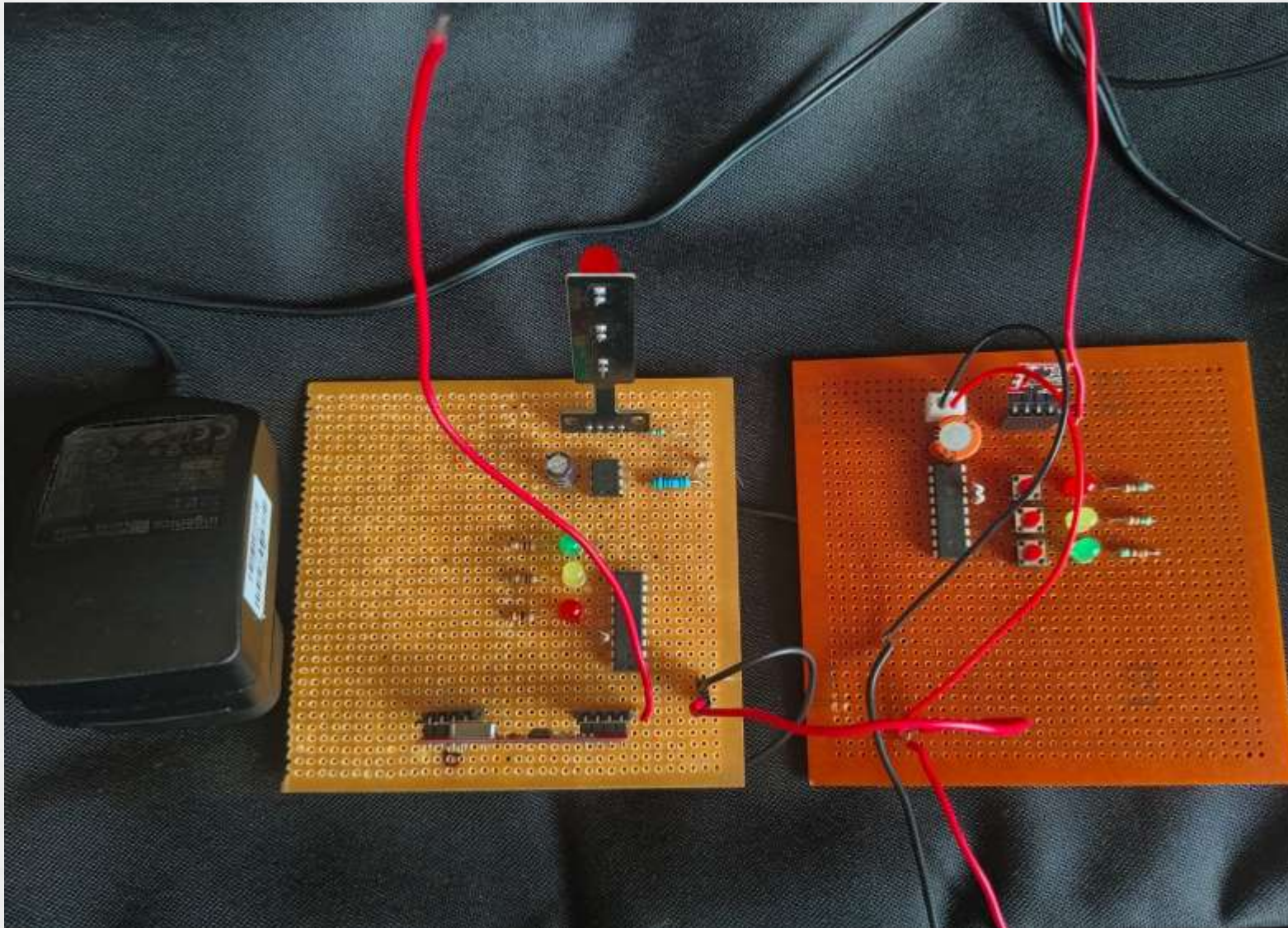
- RF Transmitter Module
- RF Receiver Module
- HT12E Encoder
- HT12D Decoder
- Traffic Signal LEDs
- Power Supply
- Breadboard and connecting wires

Working

- Emergency vehicle transmits RF signal.
- Receiver detects signal.
- Changes traffic signal to green for the specific lane.



Hardware Circuit Kit:



Results & Output:

- Emergency vehicle detection was successful.
- Traffic signal switched automatically when RF signal was detected.
- System reduced waiting time for emergency vehicles.
- Demonstrated effective priority-based traffic control.

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Conclusion & Future Scope:

Conclusion

- The proposed system successfully prioritizes emergency vehicles at traffic signals.
- It reduces response time and improves traffic management.

Future Scope

- Integration with **IoT and GPS systems**.
- Use of **AI for traffic prediction**.
- Implementation in **smart city infrastructure**.
- Mobile app integration for traffic monitoring.

Prametric Analysis :

Suppose a formula:

$$T = 1.1 \times R \times C$$

Here the parameters are:

- **R** → Resistance
- **C** → Capacitance

If you change **R** or **C**, the **time delay (T)** will change.

Real-Life Example

In your **electronics circuits**:

- Change **resistor value**
- Change **capacitor value**
- Observe the time delay changes

THANK YOU